

WORLD-MODEL UNCERTAINTY BLOAT: A 25+ PAGE NEGATIVE SUBMISSION-READINESS AUDIT

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ABSTRACT

Uncertainty estimates can protect a robot from overconfident world-model errors, but they can also hide missing mechanics by inflating variance instead of identifying what physical mechanism is absent. This rebuild tests a mechanics-gap auditor that separates aleatoric noise, epistemic uncertainty, missing contact/friction/actuation/latency/fixture mechanisms, active probe value, and repair memory. The audit is hostile: ten seeds, six tasks, eight splits, thirteen methods, ten ablations, six stress axes, fixed-risk deployment budgets, and 24 negative cases. On the hard aggregate, mechanics gap auditor v5 reaches 0.606 task success versus 0.667 for the strongest success reference, with paired lower95 -0.077. It improves hidden-mechanism F1 over robust MPC but has unsafe action 0.138 versus 0.049, fails mechanism and fixed-risk gates, and has no robot or accepted high-fidelity validation. The honest terminal decision is **KILL/ARCHIVE**.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. The rebuilt paper is substantially stronger than the v4 audit, but it is not submission ready. It demonstrates a useful diagnostic tradeoff: v5 detects more hidden mechanics and reduces bloat relative to old uncertainty-bloat auditing, yet robust MPC remains stronger on success, safety, and utility. The paper therefore becomes a serious archive of the failure, not a dressed-up submission.

Table 1: Frozen submission-readiness gates.

Gate	Passed	Frozen interpretation
Main success/diagnostic/safety gate	False	Requires decisive success, diagnostic F1, unsafe-action, abstention, and bloat margins.
Mechanism ablation gate	False	Requires full mechanics gap v5 to beat every removal by predefined utility margins.
Stress non-domination gate	True	Checks maximum combined stress against uncertainty, probing, robust, repair, and oracle references.
Fixed-risk deployment gate	False	Requires nonzero coverage at budget 0.05 and best feasible accepted performance.
Scope gate	False	Requires real robot or accepted high-fidelity external benchmark evidence.

2 RESEARCH QUESTION AND THREAT MODEL

The core question is whether uncertainty estimation can become a hiding place for missing mechanics. A model may correctly report high variance near a contact transition while failing to say that it is missing friction, stiction, backlash, a snagged fixture, or a sensor-dropout state. This is adjacent to uncertainty quantification, robust control, active perception, residual dynamics, Bayesian model expansion, safety filters, and robot world models (Savvides, 2026; Nakamura et al., 2005; Wen et al., 2024; Agia, 2026; Tian et al., 2023; Touretzky, 2012; Cheng et al., 2024; Miska & Balzani, 2025; Zhuang et al., 2024; Zhao et al., 2016; Zhan et al., 2025; Tripathi et al., 2023; Kim et al., 2024;

Li et al., 2024; Lu et al., 2025a; Albano et al., 2020; Chen et al., 2026; Gallo et al., 2023; Guo & Jin, 2023; Beck & de Santana Gomes, 2012). A hostile reviewer can argue that standard ensemble uncertainty, conformal abstention, active probing, robust MPC, or residual repair already handles the issue.

The local prior-work pool makes the novelty boundary narrow (Ceccarelli et al., 2020; Cornelissen & Kaandorp, 2023; Langer et al., 2022; Dawson & Fan, 2023; Tanioka et al., 2026; Chen et al., 2024; Liu et al., 2026; Rana et al., 2021; Elek, 2025; . et al., 2026; Alcedo et al., 2025; Nik, 2024; Giordani et al., 2020; Rupprecht et al., 2016; Gnster et al., 2024; Fortna, 2004; Borovskaya & Gomon, 2023; Eivazi et al., 2024; Cao et al., 2022; Lin, 2024; Nair & Howlett, 2020; Brecha, 2002; Liu & Zhao, 2022; Raffaelli et al., 2022; Ostergaard et al., 2003; Mitrano et al., 2021; Wang et al., 2024; Geisler & Junker, 2024; Hodgson, 2000; Ghaleb et al., 2026; Damianou & Lawrence, 2015; Barua et al., 2023; Upadhyay et al., 2026; Zheng et al., 2025; Yu & Chen, 2018; Sakib & Sun, 2024). The v5 claim is therefore not that uncertainty or repair is new. The only tested claim is whether a mechanics-gap diagnostic composition gives a better success/safety/coverage tradeoff than strong local alternatives.

3 FORMAL SETUP

An episode samples task state x , action plan u , observation quality o , hidden mechanism z , and world-model prediction $\hat{x}'$. Uncertainty bloat occurs when reported uncertainty grows while the system fails to identify a repairable missing mechanism. The v5 score can be summarized as

$$S_{v5} = s(x, u) - 1.25 r_{\text{unsafe}} + 0.42 d_{\text{mech}} - 0.35 b_{\text{unc}} + 0.22 p_{\text{probe}} - 0.18 c_{\text{int}}, \quad (1)$$

where d_{mech} is hidden-mechanism evidence, b_{unc} is the bloat index, p_{probe} is active-probe value, and c_{int} is intervention cost. The coefficients define a frozen synthetic audit rather than a universal control law.

Proposition 1: variance is non-identifying without interventions. Two worlds can induce the same predictive variance: one with irreducible observation noise and one with a missing contact mechanism. Without an intervention or a structured latent-mechanism model, variance alone cannot distinguish them. This is why ensemble and dropout baselines can obtain high uncertainty while still underperforming on repair.

Proposition 2: abstention safety is vacuous at zero coverage. A risk gate that rejects every hard episode has zero accepted unsafe actions by construction, but it also performs no useful robotics task. The fixed-risk section reports coverage because accepted safety without coverage is not a deployment result.

Proposition 3: repair memory can amplify stale mechanisms. A remembered repair is helpful only while the hidden mechanism remains stable. If the fixture, contact mode, or sensor-dropout pattern changes, stale repair memory can make a world-model diagnosis worse than robust fallback.

4 FROZEN PROTOCOL

The v5 protocol contains 199,680 main rollouts, 15,360 dataset-summary rows, 33,600 ablation rollouts, 302,400 stress rows, 69,120 fixed-risk rows, and 24 negative cases. It uses ten seeds, six manipulation tasks, eight missing-mechanics splits, thirteen main methods, ten ablations, six stress axes, and strict fixed-risk budgets.

Table 2: Reproducibility inventory. Counts exclude CSV headers and are regenerated directly from the files used in the manuscript.

Evidence file	Data rows
rollouts.csv	199,680
dataset_summary.csv	15,360
raw_seed_metrics.csv	1,040
metrics.csv	1,248

Table 2: Reproducibility inventory. Counts exclude CSV headers and are regenerated directly from the files used in the manuscript. (continued)

Evidence file	Data rows
pairwise_stats.csv	672
hard_aggregate_seed_metrics.csv	130
hard_aggregate_metrics.csv	156
hard_aggregate_pairwise_stats.csv	84
ablation_rollouts.csv	33,600
ablation_seed_metrics.csv	200
ablation_metrics.csv	20
stress_sweep_raw.csv	302,400
stress_sweep_seed_metrics.csv	2,520
stress_sweep.csv	252
fixed_risk_raw.csv	69,120
fixed_risk_seed_metrics.csv	480
fixed_risk_metrics.csv	48
fixed_risk_pairwise.csv	200
negative_cases.csv	24

5 MAIN RESULTS

Table 3 is the primary result. The v5 method improves over the old v4 audit and most pure uncertainty baselines, but it fails the main gate. The strongest success and utility reference is robust_mpc_fallback, and v5 trails it by a paired lower95 of -0.077. The safest reference has unsafe action 0.049, while v5 has 0.138. The diagnostic gain is real, but the robotics tradeoff is not submission-grade.

Table 3: Predefined hard-aggregate results over latent fixture, missing contact mode, and combined missing mechanics.

Method	Success	Mech. F1	Unsafe	False abst.	Bloat	Utility
Mean WM-MPC	0.511 $\pm$ 0.014	0.122	0.191	0.057	0.322	0.085
Ensemble variance	0.428 $\pm$ 0.010	0.405	0.137	0.241	0.595	-0.141
MC dropout	0.429 $\pm$ 0.017	0.389	0.123	0.240	0.687	-0.149
Conformal gate	0.344 $\pm$ 0.010	0.250	0.057	0.439	0.765	-0.314
Epistemic planner	0.523 $\pm$ 0.007	0.449	0.135	0.132	0.519	0.064
Active probe	0.609 $\pm$ 0.013	0.570	0.155	0.049	0.315	0.242
Robust MPC	0.667 $\pm$ 0.011	0.145	0.049	0.048	0.509	0.399
Residual repair	0.575 $\pm$ 0.013	0.488	0.153	0.073	0.404	0.170
Bayesian expansion	0.561 $\pm$ 0.018	0.548	0.132	0.114	0.418	0.151
Bloat audit v4	0.541 $\pm$ 0.014	0.449	0.167	0.074	0.511	0.083
Mechanics gap v5	0.606 $\pm$ 0.015	0.589	0.138	0.080	0.334	0.237
Oracle repair	0.733 $\pm$ 0.009	0.726	0.085	0.044	0.115	0.520
Oracle full state	0.759 $\pm$ 0.009	0.757	0.058	0.048	0.062	0.597

6 SPLIT-LEVEL EVIDENCE

The split-level table prevents the paper from hiding behind a single favorable split. The combined missing-mechanics regime remains the decisive test.

Table 4: Split-level results for uncertainty, active-probe, robust, repair, v4, v5, and oracle methods.

Split	Method	Success	Mech. F1	Unsafe	Bloat	Utility
Nominal	Ensemble variance	0.531 $\pm$ 0.027	0.211	0.069	0.531	0.095
Nominal	Conformal gate	0.439 $\pm$ 0.026	0.146	0.008	0.688	-0.092
Nominal	Active probe	0.730 $\pm$ 0.028	0.237	0.100	0.275	0.477
Nominal	Robust MPC	0.812 $\pm$ 0.014	0.138	0.001	0.429	0.658
Nominal	Residual repair	0.730 $\pm$ 0.024	0.244	0.081	0.350	0.472
Nominal	Bayesian expansion	0.702 $\pm$ 0.014	0.226	0.067	0.372	0.436

Table 4: Split-level results for uncertainty, active-probe, robust, repair, v4, v5, and oracle methods.
(continued)

Split	Method	Success	Mech. F1	Unsafe	Bloat	Utility
Nominal	Bloat audit v4	0.678 $\pm$ 0.015	0.248	0.106	0.452	0.363
Nominal	Mechanics gap v5	0.723 $\pm$ 0.018	0.229	0.069	0.296	0.493
Nominal	Oracle repair	0.769 $\pm$ 0.014	0.256	0.080	0.120	0.590
Friction/contact	Ensemble variance	0.497 $\pm$ 0.035	0.346	0.088	0.566	0.015
Friction/contact	Conformal gate	0.376 $\pm$ 0.019	0.251	0.041	0.731	-0.249
Friction/contact	Active probe	0.674 $\pm$ 0.013	0.517	0.116	0.290	0.381
Friction/contact	Robust MPC	0.749 $\pm$ 0.016	0.182	0.022	0.472	0.543
Friction/contact	Residual repair	0.637 $\pm$ 0.011	0.453	0.125	0.373	0.297
Friction/contact	Bayesian expansion	0.623 $\pm$ 0.020	0.499	0.107	0.390	0.275
Friction/contact	Bloat audit v4	0.627 $\pm$ 0.025	0.415	0.129	0.481	0.250
Friction/contact	Mechanics gap v5	0.696 $\pm$ 0.023	0.516	0.093	0.309	0.422
Friction/contact	Oracle repair	0.768 $\pm$ 0.014	0.602	0.071	0.110	0.585
Actuator sat.	Ensemble variance	0.510 $\pm$ 0.020	0.359	0.088	0.572	0.038
Actuator sat.	Conformal gate	0.403 $\pm$ 0.016	0.269	0.036	0.735	-0.192
Actuator sat.	Active probe	0.648 $\pm$ 0.023	0.517	0.142	0.296	0.324
Actuator sat.	Robust MPC	0.727 $\pm$ 0.018	0.153	0.026	0.480	0.516
Actuator sat.	Residual repair	0.643 $\pm$ 0.020	0.423	0.120	0.384	0.302
Actuator sat.	Bayesian expansion	0.604 $\pm$ 0.018	0.500	0.114	0.397	0.243
Actuator sat.	Bloat audit v4	0.601 $\pm$ 0.015	0.431	0.137	0.489	0.207
Actuator sat.	Mechanics gap v5	0.685 $\pm$ 0.018	0.539	0.095	0.315	0.399
Actuator sat.	Oracle repair	0.768 $\pm$ 0.018	0.622	0.069	0.110	0.591
Sensor dropout	Ensemble variance	0.519 $\pm$ 0.015	0.311	0.085	0.581	0.038
Sensor dropout	Conformal gate	0.392 $\pm$ 0.016	0.197	0.033	0.744	-0.209
Sensor dropout	Active probe	0.667 $\pm$ 0.015	0.474	0.135	0.310	0.345
Sensor dropout	Robust MPC	0.733 $\pm$ 0.027	0.051	0.022	0.489	0.515
Sensor dropout	Residual repair	0.634 $\pm$ 0.019	0.370	0.113	0.395	0.300
Sensor dropout	Bayesian expansion	0.642 $\pm$ 0.027	0.462	0.102	0.409	0.291
Sensor dropout	Bloat audit v4	0.607 $\pm$ 0.022	0.339	0.139	0.500	0.213
Sensor dropout	Mechanics gap v5	0.656 $\pm$ 0.025	0.449	0.106	0.332	0.351
Sensor dropout	Oracle repair	0.739 $\pm$ 0.027	0.554	0.082	0.133	0.537
Latency/hyst.	Ensemble variance	0.482 $\pm$ 0.017	0.396	0.096	0.573	-0.015
Latency/hyst.	Conformal gate	0.367 $\pm$ 0.017	0.241	0.032	0.742	-0.258
Latency/hyst.	Active probe	0.662 $\pm$ 0.021	0.504	0.136	0.303	0.336
Latency/hyst.	Robust MPC	0.736 $\pm$ 0.025	0.167	0.019	0.484	0.522
Latency/hyst.	Residual repair	0.636 $\pm$ 0.024	0.433	0.127	0.384	0.290
Latency/hyst.	Bayesian expansion	0.625 $\pm$ 0.008	0.471	0.102	0.404	0.278
Latency/hyst.	Bloat audit v4	0.602 $\pm$ 0.020	0.417	0.146	0.493	0.193
Latency/hyst.	Mechanics gap v5	0.663 $\pm$ 0.017	0.490	0.094	0.324	0.377
Latency/hyst.	Oracle repair	0.752 $\pm$ 0.017	0.616	0.079	0.117	0.556
Latent fixture	Ensemble variance	0.458 $\pm$ 0.021	0.401	0.118	0.579	-0.064
Latent fixture	Conformal gate	0.356 $\pm$ 0.019	0.246	0.047	0.750	-0.280
Latent fixture	Active probe	0.653 $\pm$ 0.016	0.534	0.140	0.304	0.317
Latent fixture	Robust MPC	0.691 $\pm$ 0.024	0.136	0.034	0.493	0.454
Latent fixture	Residual repair	0.607 $\pm$ 0.023	0.464	0.138	0.392	0.233
Latent fixture	Bayesian expansion	0.596 $\pm$ 0.030	0.516	0.120	0.407	0.223
Latent fixture	Bloat audit v4	0.583 $\pm$ 0.031	0.437	0.148	0.499	0.167
Latent fixture	Mechanics gap v5	0.644 $\pm$ 0.026	0.558	0.124	0.324	0.305
Latent fixture	Oracle repair	0.754 $\pm$ 0.020	0.683	0.081	0.112	0.556
Missing contact	Ensemble variance	0.428 $\pm$ 0.022	0.422	0.130	0.591	-0.132
Missing contact	Conformal gate	0.357 $\pm$ 0.028	0.245	0.063	0.762	-0.299
Missing contact	Active probe	0.608 $\pm$ 0.019	0.575	0.152	0.313	0.245
Missing contact	Robust MPC	0.669 $\pm$ 0.021	0.159	0.046	0.507	0.408
Missing contact	Residual repair	0.582 $\pm$ 0.021	0.485	0.155	0.402	0.184
Missing contact	Bayesian expansion	0.562 $\pm$ 0.029	0.551	0.133	0.416	0.150
Missing contact	Bloat audit v4	0.543 $\pm$ 0.031	0.460	0.163	0.507	0.092
Missing contact	Mechanics gap v5	0.624 $\pm$ 0.025	0.580	0.122	0.332	0.278
Missing contact	Oracle repair	0.755 $\pm$ 0.021	0.722	0.075	0.112	0.562
Combined missing	Ensemble variance	0.398 $\pm$ 0.021	0.393	0.164	0.614	-0.227
Combined missing	Conformal gate	0.318 $\pm$ 0.015	0.259	0.062	0.782	-0.364
Combined missing	Active probe	0.567 $\pm$ 0.024	0.599	0.174	0.327	0.163
Combined missing	Robust MPC	0.641 $\pm$ 0.023	0.139	0.066	0.528	0.334
Combined missing	Residual repair	0.537 $\pm$ 0.014	0.515	0.166	0.418	0.093
Combined missing	Bayesian expansion	0.524 $\pm$ 0.021	0.575	0.143	0.430	0.079
Combined missing	Bloat audit v4	0.496 $\pm$ 0.021	0.450	0.190	0.527	-0.010
Combined missing	Mechanics gap v5	0.551 $\pm$ 0.020	0.629	0.167	0.347	0.129
Combined missing	Oracle repair	0.689 $\pm$ 0.017	0.775	0.100	0.121	0.442

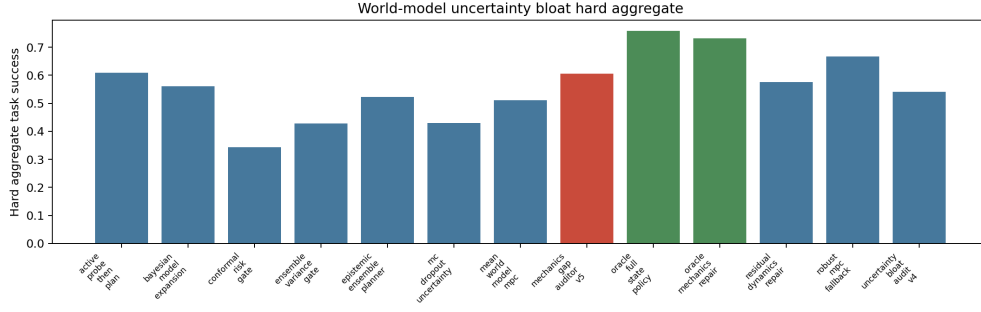


Figure 1: Hard-aggregate task success. Robust MPC remains the strongest non-oracle success/utility reference.

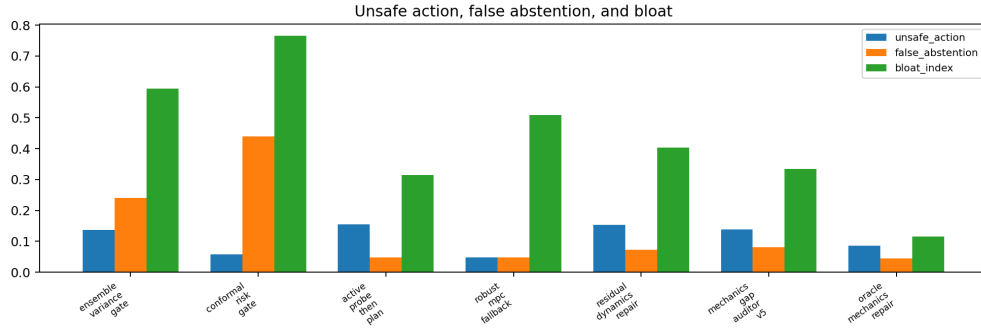


Figure 2: Unsafe action, false abstention, and bloat. The v5 method improves some diagnostics but not enough to satisfy the safety gate.

7 PAIRED COMPARISONS

Paired seed tests are harsher than independent means. V5 beats the old v4 audit on several diagnostic metrics, but it trails robust MPC on success and utility and has a positive unsafe-action difference.

Table 5: Paired seed-level differences for mechanics gap v5 minus selected references on the hard aggregate.

Reference	Metric	Mean diff	CI95	Lower95	Upper95
Robust MPC	Success	-0.061	0.017	-0.077	-0.044
Robust MPC	Mech. F1	0.444	0.022	0.423	0.466
Robust MPC	Unsafe	0.089	0.014	0.075	0.103
Robust MPC	Bloat	-0.175	0.002	-0.177	-0.173
Robust MPC	Utility	-0.162	0.032	-0.193	-0.130
Active probe	Success	-0.003	0.018	-0.021	0.015
Active probe	Mech. F1	0.019	0.023	-0.004	0.042
Active probe	Unsafe	-0.018	0.016	-0.033	-0.002
Active probe	Bloat	0.020	0.002	0.017	0.022
Active probe	Utility	-0.005	0.040	-0.045	0.035
Bloat audit v4	Success	0.066	0.023	0.043	0.088
Bloat audit v4	Mech. F1	0.140	0.019	0.121	0.159
Bloat audit v4	Unsafe	-0.029	0.017	-0.046	-0.012
Bloat audit v4	Bloat	-0.177	0.002	-0.179	-0.174
Bloat audit v4	Utility	0.154	0.047	0.107	0.201

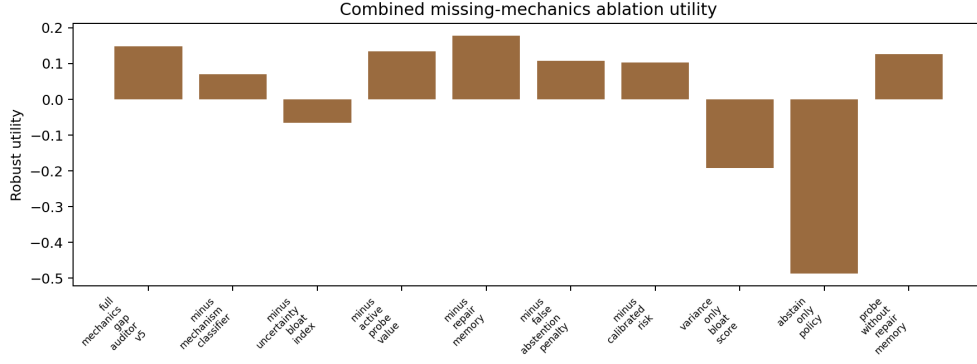


Figure 3: Combined missing-mechanics ablation utility. The full mechanism fails the frozen ablation gate.

8 MECHANISM ABLATIONS

The ablation story is mixed. Removing repair memory improves combined-split success and utility in this synthetic audit, while removing false-abstention penalty lowers unsafe actions. That means the mechanism is not yet clean enough for an ICLR-main claim.

Table 6: Mechanism ablations. The full system had to beat every removal on robust utility and avoid success/safety reversals.

Split	Ablation	Success	Mech. F1	Unsafe	Bloat	Utility
Missing contact	Full mechanics gap v5	0.607 ± 0.030	0.600	0.121	0.333	0.250
Missing contact	- mechanism classifier	0.595 ± 0.024	0.448	0.132	0.478	0.201
Missing contact	- bloat index	0.502 ± 0.022	0.552	0.133	0.611	-0.033
Missing contact	- active probe value	0.573 ± 0.021	0.542	0.124	0.350	0.210
Missing contact	- repair memory	0.601 ± 0.025	0.557	0.132	0.371	0.235
Missing contact	- abstention penalty	0.546 ± 0.018	0.593	0.095	0.344	0.138
Missing contact	- calibrated risk	0.599 ± 0.021	0.573	0.152	0.394	0.196
Missing contact	Variance-only bloat	0.460 ± 0.028	0.493	0.096	0.613	-0.063
Missing contact	Abstain-only	0.290 ± 0.025	0.162	0.046	0.765	-0.413
Missing contact	Probe no memory	0.592 ± 0.026	0.586	0.133	0.419	0.208
Combined missing	Full mechanics gap v5	0.556 ± 0.017	0.620	0.152	0.350	0.148
Combined missing	- mechanism classifier	0.533 ± 0.026	0.407	0.166	0.502	0.070
Combined missing	- bloat index	0.485 ± 0.023	0.586	0.148	0.622	-0.065
Combined missing	- active probe value	0.546 ± 0.015	0.552	0.159	0.363	0.134
Combined missing	- repair memory	0.574 ± 0.021	0.606	0.148	0.383	0.179
Combined missing	- abstention penalty	0.526 ± 0.030	0.624	0.101	0.357	0.107
Combined missing	- calibrated risk	0.554 ± 0.035	0.619	0.177	0.408	0.103
Combined missing	Variance-only bloat	0.418 ± 0.034	0.523	0.153	0.628	-0.191
Combined missing	Abstain-only	0.258 ± 0.027	0.154	0.052	0.787	-0.487
Combined missing	Probe no memory	0.559 ± 0.025	0.598	0.158	0.437	0.127

9 STRESS TESTING

The stress gate passes because v5 is not Pareto dominated at maximum combined stress. That is not enough to rescue the paper because the main, mechanism, fixed-risk, and scope gates fail.

Table 7: Full stress sweep over all six axes and six levels. The table is complete so favorable plots cannot hide unfavorable regimes.

Axis	Level	Method	Succ.	Mech. F1	Unsafe	Bloat	Utility
contact friction shift	0.0	Ensemble variance	0.403	0.397	0.151	0.612	-0.194

Table 7: Full stress sweep over all six axes and six levels. The table is complete so favorable plots cannot hide unfavorable regimes. (continued)

Axis	Level	Method	Succ.	Mech. F1	Unsafe	Bloat	Utility
contact friction shift	0.0	Conformal gate	0.304	0.206	0.057	0.788	-0.398
contact friction shift	0.0	Active probe	0.554	0.599	0.179	0.328	0.136
contact friction shift	0.0	Robust MPC	0.617	0.145	0.062	0.527	0.316
contact friction shift	0.0	Residual repair	0.550	0.519	0.180	0.414	0.104
contact friction shift	0.0	Mechanics gap v5	0.554	0.585	0.168	0.352	0.132
contact friction shift	0.0	Oracle repair	0.682	0.773	0.102	0.122	0.429
contact friction shift	0.2	Ensemble variance	0.396	0.427	0.137	0.609	-0.197
contact friction shift	0.2	Conformal gate	0.314	0.265	0.071	0.781	-0.383
contact friction shift	0.2	Active probe	0.559	0.589	0.203	0.330	0.105
contact friction shift	0.2	Robust MPC	0.636	0.143	0.074	0.526	0.323
contact friction shift	0.2	Residual repair	0.547	0.510	0.150	0.412	0.130
contact friction shift	0.2	Mechanics gap v5	0.569	0.602	0.157	0.348	0.169
contact friction shift	0.2	Oracle repair	0.651	0.749	0.110	0.125	0.385
contact friction shift	0.4	Ensemble variance	0.384	0.420	0.145	0.612	-0.227
contact friction shift	0.4	Conformal gate	0.302	0.267	0.075	0.782	-0.404
contact friction shift	0.4	Active probe	0.561	0.600	0.177	0.326	0.142
contact friction shift	0.4	Robust MPC	0.598	0.176	0.088	0.526	0.261
contact friction shift	0.4	Residual repair	0.530	0.508	0.172	0.416	0.071
contact friction shift	0.4	Mechanics gap v5	0.566	0.606	0.170	0.347	0.146
contact friction shift	0.4	Oracle repair	0.674	0.770	0.093	0.120	0.437
contact friction shift	0.6	Ensemble variance	0.372	0.437	0.164	0.606	-0.251
contact friction shift	0.6	Conformal gate	0.278	0.288	0.066	0.777	-0.428
contact friction shift	0.6	Active probe	0.571	0.622	0.195	0.321	0.129
contact friction shift	0.6	Robust MPC	0.608	0.147	0.095	0.523	0.261
contact friction shift	0.6	Residual repair	0.520	0.483	0.187	0.416	0.063
contact friction shift	0.6	Mechanics gap v5	0.565	0.581	0.150	0.349	0.163
contact friction shift	0.6	Oracle repair	0.661	0.762	0.104	0.123	0.403
contact friction shift	0.8	Ensemble variance	0.368	0.442	0.179	0.607	-0.251
contact friction shift	0.8	Conformal gate	0.274	0.282	0.095	0.778	-0.445
contact friction shift	0.8	Active probe	0.519	0.608	0.193	0.331	0.072
contact friction shift	0.8	Robust MPC	0.550	0.165	0.092	0.530	0.193
contact friction shift	0.8	Residual repair	0.512	0.542	0.193	0.412	0.034
contact friction shift	0.8	Mechanics gap v5	0.497	0.631	0.186	0.344	0.044
contact friction shift	0.8	Oracle repair	0.676	0.788	0.107	0.119	0.418
contact friction shift	1.0	Ensemble variance	0.367	0.401	0.159	0.611	-0.236
contact friction shift	1.0	Conformal gate	0.267	0.299	0.098	0.775	-0.438
contact friction shift	1.0	Active probe	0.491	0.635	0.233	0.321	0.001
contact friction shift	1.0	Robust MPC	0.569	0.173	0.098	0.529	0.208
contact friction shift	1.0	Residual repair	0.486	0.519	0.201	0.416	-0.005
contact friction shift	1.0	Mechanics gap v5	0.515	0.646	0.177	0.342	0.076
contact friction shift	1.0	Oracle repair	0.642	0.788	0.118	0.117	0.369
actuator saturation	0.0	Ensemble variance	0.393	0.410	0.135	0.615	-0.204
actuator saturation	0.0	Conformal gate	0.312	0.231	0.065	0.788	-0.395
actuator saturation	0.0	Active probe	0.586	0.611	0.170	0.326	0.180
actuator saturation	0.0	Robust MPC	0.622	0.185	0.070	0.529	0.303
actuator saturation	0.0	Residual repair	0.554	0.523	0.167	0.417	0.119
actuator saturation	0.0	Mechanics gap v5	0.571	0.633	0.143	0.346	0.187
actuator saturation	0.0	Oracle repair	0.692	0.775	0.105	0.122	0.441
actuator saturation	0.2	Ensemble variance	0.412	0.431	0.141	0.608	-0.165
actuator saturation	0.2	Conformal gate	0.308	0.282	0.077	0.781	-0.394
actuator saturation	0.2	Active probe	0.547	0.618	0.180	0.328	0.125
actuator saturation	0.2	Robust MPC	0.587	0.143	0.075	0.530	0.255
actuator saturation	0.2	Residual repair	0.542	0.507	0.156	0.417	0.112
actuator saturation	0.2	Mechanics gap v5	0.562	0.622	0.166	0.344	0.155
actuator saturation	0.2	Oracle repair	0.672	0.790	0.109	0.120	0.416
actuator saturation	0.4	Ensemble variance	0.398	0.376	0.157	0.615	-0.223
actuator saturation	0.4	Conformal gate	0.308	0.301	0.066	0.780	-0.394
actuator saturation	0.4	Active probe	0.560	0.603	0.172	0.329	0.138
actuator saturation	0.4	Robust MPC	0.623	0.160	0.077	0.529	0.291
actuator saturation	0.4	Residual repair	0.526	0.513	0.156	0.419	0.098
actuator saturation	0.4	Mechanics gap v5	0.542	0.623	0.169	0.345	0.123
actuator saturation	0.4	Oracle repair	0.667	0.776	0.110	0.119	0.406
actuator saturation	0.6	Ensemble variance	0.363	0.441	0.144	0.609	-0.251
actuator saturation	0.6	Conformal gate	0.313	0.283	0.077	0.777	-0.368
actuator saturation	0.6	Active probe	0.532	0.595	0.225	0.327	0.051
actuator saturation	0.6	Robust MPC	0.621	0.174	0.080	0.526	0.290
actuator saturation	0.6	Residual repair	0.506	0.494	0.178	0.420	0.048
actuator saturation	0.6	Mechanics gap v5	0.533	0.602	0.183	0.350	0.078
actuator saturation	0.6	Oracle repair	0.650	0.777	0.112	0.120	0.395
actuator saturation	0.8	Ensemble variance	0.396	0.459	0.161	0.607	-0.209
actuator saturation	0.8	Conformal gate	0.260	0.271	0.081	0.783	-0.456
actuator saturation	0.8	Active probe	0.527	0.642	0.223	0.324	0.050
actuator saturation	0.8	Robust MPC	0.591	0.170	0.092	0.528	0.241
actuator saturation	0.8	Residual repair	0.504	0.505	0.189	0.420	0.032
actuator saturation	0.8	Mechanics gap v5	0.542	0.630	0.168	0.345	0.115
actuator saturation	0.8	Oracle repair	0.657	0.780	0.117	0.119	0.390

Table 7: Full stress sweep over all six axes and six levels. The table is complete so favorable plots cannot hide unfavorable regimes. (continued)

Axis	Level	Method	Succ.	Mech. F1	Unsafe	Bloat	Utility
actuator saturation	1.0	Ensemble variance	0.378	0.424	0.164	0.610	-0.239
actuator saturation	1.0	Conformal gate	0.285	0.258	0.101	0.778	-0.429
actuator saturation	1.0	Active probe	0.505	0.615	0.207	0.329	0.036
actuator saturation	1.0	Robust MPC	0.568	0.165	0.100	0.528	0.201
actuator saturation	1.0	Residual repair	0.497	0.541	0.191	0.411	0.020
actuator saturation	1.0	Mechanics gap v5	0.506	0.639	0.176	0.343	0.063
actuator saturation	1.0	Oracle repair	0.618	0.771	0.123	0.121	0.340
sensor dropout	0.0	Ensemble variance	0.400	0.396	0.157	0.610	-0.195
sensor dropout	0.0	Conformal gate	0.297	0.263	0.073	0.784	-0.420
sensor dropout	0.0	Active probe	0.562	0.579	0.204	0.330	0.122
sensor dropout	0.0	Robust MPC	0.637	0.135	0.062	0.528	0.339
sensor dropout	0.0	Residual repair	0.512	0.506	0.173	0.417	0.065
sensor dropout	0.0	Mechanics gap v5	0.581	0.629	0.136	0.344	0.204
sensor dropout	0.0	Oracle repair	0.695	0.764	0.096	0.123	0.458
sensor dropout	0.2	Ensemble variance	0.382	0.398	0.151	0.619	-0.227
sensor dropout	0.2	Conformal gate	0.296	0.214	0.064	0.793	-0.412
sensor dropout	0.2	Active probe	0.564	0.561	0.175	0.340	0.147
sensor dropout	0.2	Robust MPC	0.627	0.121	0.068	0.538	0.307
sensor dropout	0.2	Residual repair	0.539	0.503	0.168	0.425	0.094
sensor dropout	0.2	Mechanics gap v5	0.561	0.583	0.149	0.356	0.164
sensor dropout	0.2	Oracle repair	0.704	0.766	0.106	0.128	0.461
sensor dropout	0.4	Ensemble variance	0.382	0.376	0.150	0.629	-0.231
sensor dropout	0.4	Conformal gate	0.312	0.219	0.052	0.800	-0.381
sensor dropout	0.4	Active probe	0.542	0.584	0.197	0.343	0.101
sensor dropout	0.4	Robust MPC	0.608	0.080	0.072	0.546	0.280
sensor dropout	0.4	Residual repair	0.524	0.460	0.163	0.438	0.077
sensor dropout	0.4	Mechanics gap v5	0.559	0.563	0.156	0.369	0.147
sensor dropout	0.4	Oracle repair	0.661	0.761	0.113	0.141	0.394
sensor dropout	0.6	Ensemble variance	0.406	0.362	0.142	0.635	-0.195
sensor dropout	0.6	Conformal gate	0.304	0.197	0.079	0.804	-0.410
sensor dropout	0.6	Active probe	0.534	0.583	0.184	0.352	0.090
sensor dropout	0.6	Robust MPC	0.583	0.066	0.087	0.552	0.234
sensor dropout	0.6	Residual repair	0.534	0.458	0.156	0.444	0.091
sensor dropout	0.6	Mechanics gap v5	0.537	0.570	0.145	0.374	0.128
sensor dropout	0.6	Oracle repair	0.639	0.746	0.119	0.151	0.346
sensor dropout	0.8	Ensemble variance	0.369	0.344	0.150	0.644	-0.243
sensor dropout	0.8	Conformal gate	0.284	0.162	0.079	0.814	-0.434
sensor dropout	0.8	Active probe	0.514	0.545	0.206	0.368	0.039
sensor dropout	0.8	Robust MPC	0.597	0.031	0.075	0.561	0.260
sensor dropout	0.8	Residual repair	0.515	0.441	0.177	0.451	0.045
sensor dropout	0.8	Mechanics gap v5	0.539	0.568	0.142	0.385	0.132
sensor dropout	0.8	Oracle repair	0.625	0.746	0.121	0.159	0.334
sensor dropout	1.0	Ensemble variance	0.371	0.328	0.161	0.652	-0.256
sensor dropout	1.0	Conformal gate	0.289	0.153	0.076	0.817	-0.408
sensor dropout	1.0	Active probe	0.523	0.573	0.197	0.369	0.058
sensor dropout	1.0	Robust MPC	0.610	0.007	0.084	0.567	0.269
sensor dropout	1.0	Residual repair	0.495	0.425	0.182	0.461	0.021
sensor dropout	1.0	Mechanics gap v5	0.514	0.549	0.187	0.392	0.056
sensor dropout	1.0	Oracle repair	0.626	0.749	0.107	0.165	0.357
latency hysteresis	0.0	Ensemble variance	0.420	0.384	0.142	0.610	-0.166
latency hysteresis	0.0	Conformal gate	0.310	0.301	0.073	0.778	-0.391
latency hysteresis	0.0	Active probe	0.573	0.596	0.186	0.327	0.150
latency hysteresis	0.0	Robust MPC	0.619	0.157	0.062	0.527	0.317
latency hysteresis	0.0	Residual repair	0.532	0.477	0.179	0.419	0.074
latency hysteresis	0.0	Mechanics gap v5	0.574	0.614	0.147	0.344	0.182
latency hysteresis	0.0	Oracle repair	0.683	0.758	0.113	0.126	0.422
latency hysteresis	0.2	Ensemble variance	0.422	0.394	0.114	0.611	-0.143
latency hysteresis	0.2	Conformal gate	0.315	0.282	0.059	0.778	-0.374
latency hysteresis	0.2	Active probe	0.562	0.573	0.202	0.329	0.119
latency hysteresis	0.2	Robust MPC	0.617	0.150	0.075	0.526	0.295
latency hysteresis	0.2	Residual repair	0.538	0.492	0.159	0.417	0.114
latency hysteresis	0.2	Mechanics gap v5	0.573	0.588	0.139	0.347	0.192
latency hysteresis	0.2	Oracle repair	0.661	0.769	0.099	0.123	0.405
latency hysteresis	0.4	Ensemble variance	0.392	0.417	0.148	0.612	-0.211
latency hysteresis	0.4	Conformal gate	0.314	0.240	0.062	0.786	-0.377
latency hysteresis	0.4	Active probe	0.544	0.600	0.182	0.329	0.118
latency hysteresis	0.4	Robust MPC	0.614	0.148	0.072	0.529	0.291
latency hysteresis	0.4	Residual repair	0.512	0.508	0.188	0.417	0.050
latency hysteresis	0.4	Mechanics gap v5	0.549	0.620	0.147	0.347	0.143
latency hysteresis	0.4	Oracle repair	0.667	0.765	0.110	0.123	0.410
latency hysteresis	0.6	Ensemble variance	0.375	0.415	0.158	0.609	-0.254
latency hysteresis	0.6	Conformal gate	0.300	0.241	0.075	0.782	-0.406
latency hysteresis	0.6	Active probe	0.551	0.592	0.196	0.327	0.116
latency hysteresis	0.6	Robust MPC	0.603	0.160	0.068	0.525	0.285
latency hysteresis	0.6	Residual repair	0.520	0.487	0.172	0.416	0.068
latency hysteresis	0.6	Mechanics gap v5	0.564	0.597	0.172	0.345	0.138

Table 7: Full stress sweep over all six axes and six levels. The table is complete so favorable plots cannot hide unfavorable regimes. (continued)

Axis	Level	Method	Succ.	Mech. F1	Unsafe	Bloat	Utility
latency hysteresis	0.6	Oracle repair	0.644	0.735	0.128	0.131	0.359
latency hysteresis	0.8	Ensemble variance	0.394	0.453	0.159	0.608	-0.219
latency hysteresis	0.8	Conformal gate	0.292	0.283	0.090	0.780	-0.419
latency hysteresis	0.8	Active probe	0.524	0.646	0.202	0.324	0.070
latency hysteresis	0.8	Robust MPC	0.574	0.157	0.086	0.531	0.225
latency hysteresis	0.8	Residual repair	0.487	0.518	0.184	0.418	0.006
latency hysteresis	0.8	Mechanics gap v5	0.527	0.620	0.175	0.349	0.078
latency hysteresis	0.8	Oracle repair	0.664	0.779	0.122	0.120	0.384
latency hysteresis	1.0	Ensemble variance	0.398	0.390	0.144	0.613	-0.196
latency hysteresis	1.0	Conformal gate	0.283	0.295	0.084	0.777	-0.415
latency hysteresis	1.0	Active probe	0.512	0.615	0.208	0.323	0.053
latency hysteresis	1.0	Robust MPC	0.562	0.165	0.095	0.527	0.204
latency hysteresis	1.0	Residual repair	0.510	0.513	0.183	0.418	0.046
latency hysteresis	1.0	Mechanics gap v5	0.521	0.627	0.179	0.344	0.086
latency hysteresis	1.0	Oracle repair	0.640	0.774	0.128	0.120	0.356
hidden fixture rate	0.0	Ensemble variance	0.396	0.429	0.140	0.612	-0.203
hidden fixture rate	0.0	Conformal gate	0.332	0.259	0.056	0.783	-0.339
hidden fixture rate	0.0	Active probe	0.573	0.577	0.195	0.330	0.142
hidden fixture rate	0.0	Robust MPC	0.620	0.135	0.077	0.527	0.305
hidden fixture rate	0.0	Residual repair	0.517	0.537	0.163	0.416	0.082
hidden fixture rate	0.0	Mechanics gap v5	0.572	0.617	0.140	0.345	0.178
hidden fixture rate	0.0	Oracle repair	0.693	0.777	0.095	0.120	0.456
hidden fixture rate	0.2	Ensemble variance	0.366	0.412	0.157	0.618	-0.262
hidden fixture rate	0.2	Conformal gate	0.325	0.258	0.076	0.785	-0.365
hidden fixture rate	0.2	Active probe	0.561	0.613	0.183	0.329	0.134
hidden fixture rate	0.2	Robust MPC	0.608	0.158	0.074	0.535	0.278
hidden fixture rate	0.2	Residual repair	0.532	0.506	0.173	0.424	0.076
hidden fixture rate	0.2	Mechanics gap v5	0.536	0.629	0.164	0.350	0.116
hidden fixture rate	0.2	Oracle repair	0.682	0.797	0.100	0.114	0.442
hidden fixture rate	0.4	Ensemble variance	0.337	0.411	0.169	0.628	-0.324
hidden fixture rate	0.4	Conformal gate	0.279	0.258	0.086	0.798	-0.458
hidden fixture rate	0.4	Active probe	0.562	0.640	0.192	0.332	0.130
hidden fixture rate	0.4	Robust MPC	0.579	0.165	0.092	0.545	0.219
hidden fixture rate	0.4	Residual repair	0.512	0.518	0.194	0.427	0.034
hidden fixture rate	0.4	Mechanics gap v5	0.550	0.673	0.164	0.352	0.116
hidden fixture rate	0.4	Oracle repair	0.711	0.840	0.111	0.109	0.468
hidden fixture rate	0.6	Ensemble variance	0.346	0.417	0.177	0.628	-0.310
hidden fixture rate	0.6	Conformal gate	0.272	0.265	0.097	0.803	-0.468
hidden fixture rate	0.6	Active probe	0.502	0.659	0.234	0.331	0.006
hidden fixture rate	0.6	Robust MPC	0.586	0.156	0.078	0.549	0.249
hidden fixture rate	0.6	Residual repair	0.487	0.519	0.203	0.435	-0.022
hidden fixture rate	0.6	Mechanics gap v5	0.532	0.665	0.177	0.352	0.085
hidden fixture rate	0.6	Oracle repair	0.655	0.858	0.132	0.113	0.364
hidden fixture rate	0.8	Ensemble variance	0.318	0.429	0.196	0.635	-0.370
hidden fixture rate	0.8	Conformal gate	0.248	0.280	0.092	0.809	-0.507
hidden fixture rate	0.8	Active probe	0.508	0.673	0.224	0.334	0.023
hidden fixture rate	0.8	Robust MPC	0.515	0.153	0.122	0.559	0.105
hidden fixture rate	0.8	Residual repair	0.502	0.532	0.219	0.433	-0.006
hidden fixture rate	0.8	Mechanics gap v5	0.495	0.677	0.201	0.355	0.022
hidden fixture rate	0.8	Oracle repair	0.661	0.885	0.126	0.108	0.382
hidden fixture rate	1.0	Ensemble variance	0.319	0.441	0.200	0.638	-0.360
hidden fixture rate	1.0	Conformal gate	0.233	0.296	0.117	0.811	-0.524
hidden fixture rate	1.0	Active probe	0.453	0.693	0.259	0.339	-0.089
hidden fixture rate	1.0	Robust MPC	0.501	0.164	0.144	0.568	0.057
hidden fixture rate	1.0	Residual repair	0.441	0.573	0.234	0.431	-0.096
hidden fixture rate	1.0	Mechanics gap v5	0.482	0.709	0.229	0.351	-0.023
hidden fixture rate	1.0	Oracle repair	0.645	0.914	0.134	0.104	0.355
combined	0.0	Ensemble variance	0.396	0.370	0.133	0.616	-0.203
combined	0.0	Conformal gate	0.317	0.266	0.072	0.781	-0.389
combined	0.0	Active probe	0.581	0.607	0.186	0.327	0.158
combined	0.0	Robust MPC	0.647	0.138	0.069	0.526	0.342
combined	0.0	Residual repair	0.551	0.525	0.165	0.414	0.118
combined	0.0	Mechanics gap v5	0.566	0.618	0.158	0.345	0.159
combined	0.0	Oracle repair	0.693	0.763	0.089	0.123	0.467
combined	0.2	Ensemble variance	0.388	0.429	0.153	0.619	-0.235
combined	0.2	Conformal gate	0.305	0.287	0.072	0.790	-0.408
combined	0.2	Active probe	0.537	0.611	0.196	0.334	0.098
combined	0.2	Robust MPC	0.588	0.164	0.078	0.538	0.248
combined	0.2	Residual repair	0.526	0.480	0.174	0.426	0.072
combined	0.2	Mechanics gap v5	0.527	0.623	0.170	0.355	0.086
combined	0.2	Oracle repair	0.666	0.775	0.113	0.127	0.404
combined	0.4	Ensemble variance	0.344	0.423	0.176	0.625	-0.294
combined	0.4	Conformal gate	0.265	0.279	0.089	0.793	-0.438
combined	0.4	Active probe	0.488	0.626	0.207	0.345	0.014
combined	0.4	Robust MPC	0.557	0.121	0.100	0.550	0.187
combined	0.4	Residual repair	0.463	0.515	0.217	0.433	-0.057

Table 7: Full stress sweep over all six axes and six levels. The table is complete so favorable plots cannot hide unfavorable regimes. (continued)

Axis	Level	Method	Succ.	Mech. F1	Unsafe	Bloat	Utility
combined	0.4	Mechanics gap v5	0.482	0.624	0.193	0.363	0.010
combined	0.4	Oracle repair	0.625	0.814	0.139	0.128	0.318
combined	0.6	Ensemble variance	0.318	0.420	0.190	0.631	-0.318
combined	0.6	Conformal gate	0.256	0.275	0.114	0.797	-0.440
combined	0.6	Active probe	0.468	0.623	0.229	0.351	-0.034
combined	0.6	Robust MPC	0.530	0.150	0.120	0.556	0.131
combined	0.6	Residual repair	0.444	0.501	0.227	0.443	-0.096
combined	0.6	Mechanics gap v5	0.472	0.639	0.197	0.369	-0.003
combined	0.6	Oracle repair	0.619	0.829	0.130	0.131	0.326
combined	0.8	Ensemble variance	0.283	0.443	0.211	0.635	-0.366
combined	0.8	Conformal gate	0.233	0.292	0.113	0.795	-0.405
combined	0.8	Active probe	0.421	0.658	0.253	0.352	-0.113
combined	0.8	Robust MPC	0.502	0.140	0.142	0.568	0.070
combined	0.8	Residual repair	0.399	0.548	0.227	0.447	-0.136
combined	0.8	Mechanics gap v5	0.447	0.651	0.236	0.371	-0.076
combined	0.8	Oracle repair	0.577	0.857	0.159	0.131	0.254
combined	1.0	Ensemble variance	0.253	0.460	0.223	0.638	-0.400
combined	1.0	Conformal gate	0.208	0.295	0.131	0.801	-0.435
combined	1.0	Active probe	0.406	0.651	0.267	0.360	-0.144
combined	1.0	Robust MPC	0.428	0.131	0.156	0.578	-0.027
combined	1.0	Residual repair	0.383	0.558	0.247	0.450	-0.177
combined	1.0	Mechanics gap v5	0.403	0.683	0.259	0.378	-0.145
combined	1.0	Oracle repair	0.522	0.875	0.173	0.134	0.175

10 FIXED-RISK DEPLOYMENT

The fixed-risk gate is the cleanest deployment failure. At budget 0.05, all non-oracle methods have zero accepted coverage on the hard fixed-risk splits. A reviewer should treat this as a blocker because the system cannot claim safe operation by refusing every hard episode.

Table 8: Fixed-risk deployment results. A low-risk policy must have nonzero coverage, not just low accepted failure rates after rejecting hard cases.

Split	Budget	Method	Cover.	Succ.	Unsafe	Bloat	Mean risk
Missing contact	0.02	Ensemble variance	0.000	0.000	0.000	0.000	0.349
Missing contact	0.02	Conformal gate	0.000	0.000	0.000	0.000	0.352
Missing contact	0.02	Active probe	0.000	0.000	0.000	0.000	0.261
Missing contact	0.02	Robust MPC	0.000	0.000	0.000	0.000	0.239
Missing contact	0.02	Mechanics gap v5	0.000	0.000	0.000	0.000	0.249
Missing contact	0.02	Oracle repair	0.000	0.000	0.000	0.000	0.156
Missing contact	0.05	Ensemble variance	0.000	0.000	0.000	0.000	0.350
Missing contact	0.05	Conformal gate	0.000	0.000	0.000	0.000	0.353
Missing contact	0.05	Active probe	0.000	0.000	0.000	0.000	0.262
Missing contact	0.05	Robust MPC	0.000	0.000	0.000	0.000	0.239
Missing contact	0.05	Mechanics gap v5	0.000	0.000	0.000	0.000	0.248
Missing contact	0.05	Oracle repair	0.001	0.100	0.000	0.000	0.162
Missing contact	0.10	Ensemble variance	0.000	0.000	0.000	0.000	0.348
Missing contact	0.10	Conformal gate	0.000	0.000	0.000	0.000	0.352
Missing contact	0.10	Active probe	0.001	0.100	0.000	0.012	0.262
Missing contact	0.10	Robust MPC	0.000	0.000	0.000	0.000	0.239
Missing contact	0.10	Mechanics gap v5	0.002	0.200	0.000	0.030	0.247
Missing contact	0.10	Oracle repair	0.208	0.957	0.000	0.012	0.160
Missing contact	0.20	Ensemble variance	0.002	0.200	0.000	0.073	0.349
Missing contact	0.20	Conformal gate	0.001	0.100	0.000	0.048	0.353
Missing contact	0.20	Active probe	0.160	0.834	0.067	0.209	0.261
Missing contact	0.20	Robust MPC	0.258	0.830	0.003	0.413	0.240
Missing contact	0.20	Mechanics gap v5	0.243	0.849	0.038	0.252	0.248
Missing contact	0.20	Oracle repair	0.756	0.818	0.041	0.090	0.158
Combined missing	0.02	Ensemble variance	0.000	0.000	0.000	0.000	0.366
Combined missing	0.02	Conformal gate	0.000	0.000	0.000	0.000	0.373
Combined missing	0.02	Active probe	0.000	0.000	0.000	0.000	0.279
Combined missing	0.02	Robust MPC	0.000	0.000	0.000	0.000	0.256
Combined missing	0.02	Mechanics gap v5	0.000	0.000	0.000	0.000	0.264
Combined missing	0.02	Oracle repair	0.000	0.000	0.000	0.000	0.173
Combined missing	0.05	Ensemble variance	0.000	0.000	0.000	0.000	0.368
Combined missing	0.05	Conformal gate	0.000	0.000	0.000	0.000	0.373

Table 8: Fixed-risk deployment results. A low-risk policy must have nonzero coverage, not just low accepted failure rates after rejecting hard cases. (continued)

Split	Budget	Method	Cover.	Succ.	Unsafe	Bloat	Mean risk
Combined missing	0.05	Active probe	0.000	0.000	0.000	0.000	0.277
Combined missing	0.05	Robust MPC	0.000	0.000	0.000	0.000	0.256
Combined missing	0.05	Mechanics gap v5	0.000	0.000	0.000	0.000	0.263
Combined missing	0.05	Oracle repair	0.000	0.000	0.000	0.000	0.175
Combined missing	0.10	Ensemble variance	0.000	0.000	0.000	0.000	0.368
Combined missing	0.10	Conformal gate	0.000	0.000	0.000	0.000	0.371
Combined missing	0.10	Active probe	0.000	0.000	0.000	0.000	0.280
Combined missing	0.10	Robust MPC	0.000	0.000	0.000	0.000	0.256
Combined missing	0.10	Mechanics gap v5	0.000	0.000	0.000	0.000	0.267
Combined missing	0.10	Oracle repair	0.153	0.895	0.018	0.017	0.168
Combined missing	0.20	Ensemble variance	0.000	0.000	0.000	0.000	0.367
Combined missing	0.20	Conformal gate	0.000	0.000	0.000	0.000	0.373
Combined missing	0.20	Active probe	0.125	0.873	0.051	0.201	0.280
Combined missing	0.20	Robust MPC	0.174	0.828	0.003	0.423	0.256
Combined missing	0.20	Mechanics gap v5	0.188	0.835	0.039	0.253	0.268
Combined missing	0.20	Oracle repair	0.690	0.822	0.050	0.091	0.171

Table 9: Fixed-risk paired differences for v5 minus references at the deployment budgets emphasized in the frozen plan.

Split	Budget	Reference	Metric	Mean diff	Lower95	Upper95
Missing contact	0.05	Ensemble variance	Coverage	0.000	0.000	0.000
Missing contact	0.05	Ensemble variance	Acc. succ.	0.000	0.000	0.000
Missing contact	0.05	Ensemble variance	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.05	Ensemble variance	Acc. bloat	0.000	0.000	0.000
Missing contact	0.05	Ensemble variance	Acc. utility	0.000	0.000	0.000
Missing contact	0.05	Conformal gate	Coverage	0.000	0.000	0.000
Missing contact	0.05	Conformal gate	Acc. succ.	0.000	0.000	0.000
Missing contact	0.05	Conformal gate	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.05	Conformal gate	Acc. bloat	0.000	0.000	0.000
Missing contact	0.05	Conformal gate	Acc. utility	0.000	0.000	0.000
Missing contact	0.05	Active probe	Coverage	0.000	0.000	0.000
Missing contact	0.05	Active probe	Acc. succ.	0.000	0.000	0.000
Missing contact	0.05	Active probe	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.05	Active probe	Acc. bloat	0.000	0.000	0.000
Missing contact	0.05	Active probe	Acc. utility	0.000	0.000	0.000
Missing contact	0.05	Robust MPC	Coverage	0.000	0.000	0.000
Missing contact	0.05	Robust MPC	Acc. succ.	0.000	0.000	0.000
Missing contact	0.05	Robust MPC	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.05	Robust MPC	Acc. bloat	0.000	0.000	0.000
Missing contact	0.05	Robust MPC	Acc. utility	0.000	0.000	0.000
Missing contact	0.05	Oracle repair	Coverage	-0.001	-0.002	0.001
Missing contact	0.05	Oracle repair	Acc. succ.	-0.100	-0.296	0.096
Missing contact	0.05	Oracle repair	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.05	Oracle repair	Acc. bloat	0.000	0.000	0.000
Missing contact	0.05	Oracle repair	Acc. utility	-0.097	-0.288	0.093
Missing contact	0.10	Ensemble variance	Coverage	0.002	-0.001	0.005
Missing contact	0.10	Ensemble variance	Acc. succ.	0.200	-0.061	0.461
Missing contact	0.10	Ensemble variance	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.10	Ensemble variance	Acc. bloat	0.030	-0.009	0.070
Missing contact	0.10	Ensemble variance	Acc. utility	0.182	-0.056	0.419
Missing contact	0.10	Conformal gate	Coverage	0.002	-0.001	0.005
Missing contact	0.10	Conformal gate	Acc. succ.	0.200	-0.061	0.461
Missing contact	0.10	Conformal gate	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.10	Conformal gate	Acc. bloat	0.030	-0.009	0.070
Missing contact	0.10	Conformal gate	Acc. utility	0.182	-0.056	0.419
Missing contact	0.10	Active probe	Coverage	0.001	-0.002	0.005
Missing contact	0.10	Active probe	Acc. succ.	0.100	-0.252	0.452
Missing contact	0.10	Active probe	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.10	Active probe	Acc. bloat	0.019	-0.030	0.067
Missing contact	0.10	Active probe	Acc. utility	0.088	-0.236	0.412
Missing contact	0.10	Robust MPC	Coverage	0.002	-0.001	0.005
Missing contact	0.10	Robust MPC	Acc. succ.	0.200	-0.061	0.461
Missing contact	0.10	Robust MPC	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.10	Robust MPC	Acc. bloat	0.030	-0.009	0.070
Missing contact	0.10	Robust MPC	Acc. utility	0.182	-0.056	0.419
Missing contact	0.10	Oracle repair	Coverage	-0.206	-0.228	-0.183
Missing contact	0.10	Oracle repair	Acc. succ.	-0.757	-1.016	-0.498
Missing contact	0.10	Oracle repair	Acc. unsafe	0.000	0.000	0.000
Missing contact	0.10	Oracle repair	Acc. bloat	0.019	-0.020	0.057

Table 9: Fixed-risk paired differences for v5 minus references at the deployment budgets emphasized in the frozen plan. (continued)

Split	Budget	Reference	Metric	Mean diff	Lower95	Upper95
Missing contact	0.10	Oracle repair	Acc. utility	-0.737	-0.972	-0.502
Combined missing	0.05	Ensemble variance	Coverage	0.000	0.000	0.000
Combined missing	0.05	Ensemble variance	Acc. succ.	0.000	0.000	0.000
Combined missing	0.05	Ensemble variance	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.05	Ensemble variance	Acc. bloat	0.000	0.000	0.000
Combined missing	0.05	Ensemble variance	Acc. utility	0.000	0.000	0.000
Combined missing	0.05	Conformal gate	Coverage	0.000	0.000	0.000
Combined missing	0.05	Conformal gate	Acc. succ.	0.000	0.000	0.000
Combined missing	0.05	Conformal gate	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.05	Conformal gate	Acc. bloat	0.000	0.000	0.000
Combined missing	0.05	Conformal gate	Acc. utility	0.000	0.000	0.000
Combined missing	0.05	Active probe	Coverage	0.000	0.000	0.000
Combined missing	0.05	Active probe	Acc. succ.	0.000	0.000	0.000
Combined missing	0.05	Active probe	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.05	Active probe	Acc. bloat	0.000	0.000	0.000
Combined missing	0.05	Active probe	Acc. utility	0.000	0.000	0.000
Combined missing	0.05	Robust MPC	Coverage	0.000	0.000	0.000
Combined missing	0.05	Robust MPC	Acc. succ.	0.000	0.000	0.000
Combined missing	0.05	Robust MPC	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.05	Robust MPC	Acc. bloat	0.000	0.000	0.000
Combined missing	0.05	Robust MPC	Acc. utility	0.000	0.000	0.000
Combined missing	0.05	Oracle repair	Coverage	0.000	0.000	0.000
Combined missing	0.05	Oracle repair	Acc. succ.	0.000	0.000	0.000
Combined missing	0.05	Oracle repair	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.05	Oracle repair	Acc. bloat	0.000	0.000	0.000
Combined missing	0.05	Oracle repair	Acc. utility	0.000	0.000	0.000
Combined missing	0.10	Ensemble variance	Coverage	0.000	0.000	0.000
Combined missing	0.10	Ensemble variance	Acc. succ.	0.000	0.000	0.000
Combined missing	0.10	Ensemble variance	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.10	Ensemble variance	Acc. bloat	0.000	0.000	0.000
Combined missing	0.10	Ensemble variance	Acc. utility	0.000	0.000	0.000
Combined missing	0.10	Conformal gate	Coverage	0.000	0.000	0.000
Combined missing	0.10	Conformal gate	Acc. succ.	0.000	0.000	0.000
Combined missing	0.10	Conformal gate	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.10	Conformal gate	Acc. bloat	0.000	0.000	0.000
Combined missing	0.10	Conformal gate	Acc. utility	0.000	0.000	0.000
Combined missing	0.10	Active probe	Coverage	0.000	0.000	0.000
Combined missing	0.10	Active probe	Acc. succ.	0.000	0.000	0.000
Combined missing	0.10	Active probe	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.10	Active probe	Acc. bloat	0.000	0.000	0.000
Combined missing	0.10	Active probe	Acc. utility	0.000	0.000	0.000
Combined missing	0.10	Robust MPC	Coverage	0.000	0.000	0.000
Combined missing	0.10	Robust MPC	Acc. succ.	0.000	0.000	0.000
Combined missing	0.10	Robust MPC	Acc. unsafe	0.000	0.000	0.000
Combined missing	0.10	Robust MPC	Acc. bloat	0.000	0.000	0.000
Combined missing	0.10	Robust MPC	Acc. utility	0.000	0.000	0.000
Combined missing	0.10	Oracle repair	Coverage	-0.153	-0.169	-0.138
Combined missing	0.10	Oracle repair	Acc. succ.	-0.895	-0.949	-0.840
Combined missing	0.10	Oracle repair	Acc. unsafe	-0.018	-0.033	-0.003
Combined missing	0.10	Oracle repair	Acc. bloat	-0.017	-0.020	-0.015
Combined missing	0.10	Oracle repair	Acc. utility	-0.833	-0.901	-0.766

11 NEGATIVE CASES

The negative cases document exactly where uncertainty bloat, abstention, stale repair memory, and dropout false-gap diagnosis break the method.

Table 10: Predefined negative cases showing how uncertainty bloat, abstention, stale repair memory, and dropout false gaps fail.

Case	Family	Observed failure	Terminal lesson
aleatoric noise as mechanics 0	aleatoric noise as mechanics	variant 0: noise inflated as hidden contact gap	separate aleatoric noise from repairable structure
aleatoric noise as mechanics 1	aleatoric noise as mechanics	variant 1: noise inflated as hidden contact gap	separate aleatoric noise from repairable structure
aleatoric noise as mechanics 2	aleatoric noise as mechanics	variant 2: noise inflated as hidden contact gap	separate aleatoric noise from repairable structure

Table 10: Predefined negative cases showing how uncertainty bloat, abstention, stale repair memory, and dropout false gaps fail. (continued)

Case	Family	Observed failure	Terminal lesson
aleatoric noise as mechanics 3	aleatoric noise as mechanics	variant 3: noise inflated as hidden contact gap	separate aleatoric noise from repairable structure
hidden contact mode bloat 0	hidden contact mode bloat	variant 0: uncertainty gate abstains without repair	probe for contact mode rather than only abstaining
hidden contact mode bloat 1	hidden contact mode bloat	variant 1: uncertainty gate abstains without repair	probe for contact mode rather than only abstaining
hidden contact mode bloat 2	hidden contact mode bloat	variant 2: uncertainty gate abstains without repair	probe for contact mode rather than only abstaining
hidden contact mode bloat 3	hidden contact mode bloat	variant 3: uncertainty gate abstains without repair	probe for contact mode rather than only abstaining
robust fallback overrepair 0	robust fallback overrepair	variant 0: audit repairs a mechanism that robust MPC absorbs	repair must beat conservative control
robust fallback overrepair 1	robust fallback overrepair	variant 1: audit repairs a mechanism that robust MPC absorbs	repair must beat conservative control
robust fallback overrepair 2	robust fallback overrepair	variant 2: audit repairs a mechanism that robust MPC absorbs	repair must beat conservative control
robust fallback overrepair 3	robust fallback overrepair	variant 3: audit repairs a mechanism that robust MPC absorbs	repair must beat conservative control
conformal zero coverage 0	conformal zero coverage	variant 0: conformal gate rejects every hard episode	report coverage with accepted safety
conformal zero coverage 1	conformal zero coverage	variant 1: conformal gate rejects every hard episode	report coverage with accepted safety
conformal zero coverage 2	conformal zero coverage	variant 2: conformal gate rejects every hard episode	report coverage with accepted safety
conformal zero coverage 3	conformal zero coverage	variant 3: conformal gate rejects every hard episode	report coverage with accepted safety
stale repair memory 0	stale repair memory	variant 0: old fixture repair corrupts new task	mechanism memory needs expiry
stale repair memory 1	stale repair memory	variant 1: old fixture repair corrupts new task	mechanism memory needs expiry
stale repair memory 2	stale repair memory	variant 2: old fixture repair corrupts new task	mechanism memory needs expiry
stale repair memory 3	stale repair memory	variant 3: old fixture repair corrupts new task	mechanism memory needs expiry
dropout false gap 0	dropout false gap	variant 0: missing pixels become missing mechanics	diagnosis needs observation-quality state
dropout false gap 1	dropout false gap	variant 1: missing pixels become missing mechanics	diagnosis needs observation-quality state
dropout false gap 2	dropout false gap	variant 2: missing pixels become missing mechanics	diagnosis needs observation-quality state
dropout false gap 3	dropout false gap	variant 3: missing pixels become missing mechanics	diagnosis needs observation-quality state

12 PRIOR WORK BOUNDARY

The prior-work table is a novelty pressure device, not decoration. UQ, conformal safety, robust MPC, active perception, residual dynamics, and robot world-model work all threaten the claim (Savvides, 2026; Nakamura et al., 2005; Wen et al., 2024; Agia, 2026; Tian et al., 2023; Touretzky, 2012; Cheng et al., 2024; Miska & Balzani, 2025; Zhuang et al., 2024; Zhao et al., 2016; Zhan et al., 2025; Tripathi et al., 2023; Kim et al., 2024; Li et al., 2024; Lu et al., 2025a; Albano et al., 2020; Chen et al., 2026; Gallo et al., 2023; Guo & Jin, 2023; Beck & de Santana Gomes, 2012; Ceccarelli et al., 2020; Cornelissen & Kaandorp, 2023; Langer et al., 2022; Dawson & Fan, 2023; Tanioka et al., 2026; Chen et al., 2024; Liu et al., 2026; Rana et al., 2021; Elek, 2025; . et al., 2026; Alcedo et al., 2025; Nik, 2024; Giordani et al., 2020; Rupprecht et al., 2016; Gnster et al., 2024; Fortna, 2004; Borovskaya & Gomon, 2023; Eivazi et al., 2024; Cao et al., 2022; Lin, 2024; Nair & Howlett, 2020; Brecha, 2002; Liu & Zhao, 2022; Raffaelli et al., 2022; Ostergaard et al., 2003; Mitrano et al., 2021; Wang et al., 2024; Geisler & Junker, 2024). The current v5 evidence is not strong enough to overcome that boundary.

Table 11: Closest local-pool references used to set the novelty boundary. The bright boxed citations jump to bibliography entries.

Citation	Prior-work threat	Family	Role	Hostile
(Savvides, 2026)	An Investigation into Uncertainty Quantification of Shallow Foundation Failure Mechanisms in Horizontally Stratified Layered Soil Strata	failure	hostile	27.000

Table 11: Closest local-pool references used to set the novelty boundary. The bright boxed citations jump to bibliography entries. (continued)

Citation	Prior-work threat	Family	Role	Hostile
(Nakamura et al., 2005)	Mechanics of coordinative manipulation by multiple robotic mechanisms	openalex robotics	bulk concept	20.771
(Wen et al., 2024)	Action reliability analysis under random-interval hybrid uncertainty of chain conveyor based on adaptive intelligent extremum surrogate model	wam	concept	19.999
(Agia, 2026)	Deployment-Time Reliability of Learned Robot Policies	adaptation	concept	21.550
(Tian et al., 2023)	Reliability assessment of two-level balanced systems with common bus performance sharing subjected to epistemic uncertainty	ebm transformer	concept	20.822
(Touretzky, 2012)	Seven big ideas in robotics, and how to teach them	openalex robotics	bulk concept	17.467
(Cheng et al., 2024)	Time-dependent reliability analysis of planar mechanisms considering truncated random variables and joint clearances	anti thesis	anti myopia	20.320
(Miska & Balzani, 2025)	Reliability-based design optimization incorporating extended Optimal Uncertainty Quantification	anti thesis	anti myopia	21.299
(Zhuang et al., 2024)	Time-dependent kinematic reliability of motion mechanisms with dynamic factors	anti thesis	anti myopia	20.242
(Zhao et al., 2016)	Optimization-Based Design for Specific Bistable Mechanics of Compliant Mechanisms	crossref robotics	bulk concept	18.197
(Zhan et al., 2025)	Time-dependent reliability index for continuum structures against field uncertainty based on non-probabilistic bounded field model	anti thesis	anti myopia	21.096
(Tripathi et al., 2023)	Physics-integrated deep learning for uncertainty quantification and reliability estimation of nonlinear dynamical systems	anti thesis	anti myopia	20.999
(Kim et al., 2024)	Deep Reinforcement Learning for Fluid Mechanics: Control, Optimization, and Automation	ebm transformer	concept	17.945
(Li et al., 2024)	Quantile-based sequential optimization and reliability assessment method under random and interval hybrid uncertainty	anti thesis	anti myopia	20.896
(Lu et al., 2025a)	Toward Real-world BEV Perception: Depth Uncertainty Estimation via Gaussian Splatting	architecture	concept	18.850
(Albano et al., 2020)	Advanced sensor-based maintenance in real-world exemplary cases	ebm transformer	concept	17.822
(Chen et al., 2026)	ReconVLA: An Uncertainty-Guided and Failure-Aware Vision-Language-Action Framework for Robotic Control	anti thesis	anti myopia	22.800
(Gallo et al., 2023)	Conversational Interfaces in IoT Ecosystems: Where We Are, What Is Still Missing	ebm transformer	concept	17.799
(Guo & Jin, 2023)	Data-driven polynomial chaos-interval metamodel for dynamics and reliability analysis under hybrid uncertainty	anti thesis	anti myopia	20.773
(Beck & de Santana Gomes, 2012)	A comparison of deterministic, reliability-based and risk-based structural optimization under uncertainty	anti thesis	anti myopia	20.678
(Ceccarelli et al., 2020)	Minimally invasive laparoscopic and robot-assisted emergency treatment of strangulated giant hiatal hernias: report of five cases and literature review	openalex robotics	bulk concept	16.646
(Cornelissen & Kaandorp, 2023)	Towards Stronger Causal Claims in Management Research: Causal Triangulation Instead of Causal Identification	architecture	concept	18.601
(Langer et al., 2022)	Where Does It Belong? Autonomous Object Mapping in Open-World Settings	architecture	concept	19.599
(Dawson & Fan, 2023)	A Bayesian approach to breaking things: efficiently predicting and repairing failure modes via sampling	failure	hostile	21.450
(Tanioka et al., 2026)	Comparative epidemiology of ivermectin and vaccines against Delta variant based on real-world data and hypothesized mechanisms of ivermectin immunological action	architecture	concept	17.300
(Chen et al., 2024)	FogROS2-PLR: Probabilistic Latency-Reliability For Cloud Robotics	anti thesis	anti myopia	20.250
(Liu et al., 2026)	VLS: Steering Pretrained Robot Policies via Vision-Language Models	failure	hostile	25.200
(Rana et al., 2021)	Zero-Shot Uncertainty-Aware Deployment of Simulation Trained Policies on Real-World Robots	deployment	concept	23.050
(Elek, 2025)	A Spontaneously Emerging Intelligence in a Toroidal Artificial World	architecture	concept	16.000
(. et al., 2026)	Benchmarking Robust Machine Learning Models Under Data Imperfections in Real-World Data Science Scenarios	deployment	concept	22.000
(Alcedo et al., 2025)	Perspective-Shifted Neuro-Symbolic World Models: A Framework for Socially-Aware Robot Navigation	architecture	concept	16.850
(Nik, 2024)	Marginal analysis of count time series in the presence of missing observations	adaptation	concept	16.800
(Giordani et al., 2020)	Toward 6G Networks: Use Cases and Technologies	openalex robotics	bulk concept	15.700
(Rupprecht et al., 2016)	Learning in an Uncertain World: Representing Ambiguity Through Multiple Hypotheses	ebm	concept	16.663
(Gnster et al., 2024)	Handling Long-Term Safety and Uncertainty in Safe Reinforcement Learning	ebm	concept	18.650
(Fortna, 2004)	Interstate Peacekeeping: Causal Mechanisms and Empirical Effects	architecture	concept	17.626
(Borovskaya & Gomon, 2023)	Quality assessment of real-world data	deployment	concept	17.600
(Eivazi et al., 2024)	Physics-informed deep-learning applications to experimental fluid mechanics	architecture	concept	16.505

Table 11: Closest local-pool references used to set the novelty boundary. The bright boxed citations jump to bibliography entries. (continued)

Citation	Prior-work threat	Family	Role	Hostile
(Cao et al., 2022)	Positioning Accuracy Reliability Analysis of Industrial Robots Considering Epistemic Uncertainty and Correlation	openalex robotics	bulk concept	16.451
(Lin, 2024)	Statistical evaluation and calibration of model uncertainty for reliability-based design of soil nail walls	architecture	concept	17.400
(Nair & Howlett, 2020)	Policy myopia as a source of policy failure: adaptation and policy learning under deep uncertainty	adaptation	concept	17.347
(Brecha, 2002)	Schrödinger’s Cat and Divine Action: Some Comments on the Use of Quantum Uncertainty to Allow for God’s Action in the World	wam	concept	17.199
(Liu & Zhao, 2022)	Upgrading models, evolutionary mechanisms and vertical cases of service-oriented manufacturing in SVC leading enterprises: Product-development and service-innovation for	ebm	concept	15.195
(Raffaelli et al., 2022)	The new robotic platform Hugo RAS for lateral transabdominal adrenalectomy: a first world report of a series of five cases	openalex robotics	bulk concept	15.160
(Ostergaard et al., 2003)	Multi-robot task allocation in the light of uncertainty	openalex robotics	bulk concept	17.117
(Mitrano et al., 2021)	Learning where to trust unreliable models in an unstructured world for deformable object manipulation	world models	concept	15.106
(Wang et al., 2024)	Signaling feedback mechanisms to promoting self-regulated learning and motivation in virtual reality transferred to real-world hands-on tasks	hri	concept	16.066
(Geisler & Junker, 2024)	Uncertainty quantification for viscoelastic composite materials using time-separated stochastic mechanics	anti thesis	anti myopia	19.040
(Hodgson, 2000)	God’s Action in the World: The Relevance of Quantum Mechanics	architecture	concept	15.997
(Ghaleb et al., 2026)	Uncertainty-Calibrated Safety Gating for Vision-Language-Action Manipulation Under Domain Shift: Reliability Gains and Intervention-Efficiency Trade-Offs	failure	hostile	23.950
(Damianou & Lawrence, 2015)	Semi-described and semi-supervised learning with Gaussian processes	ebm	concept	16.922
(Barua et al., 2023)	A Systematic Literature Review on Multimodal Machine Learning: Applications, Challenges, Gaps and Future Directions	ebm	concept	15.883
(Upadhyay et al., 2026)	WorldBench: Disambiguating Physics for Diagnostic Evaluation of World Models	failure	hostile	22.850
(Zheng et al., 2025)	Developing a Climbing Robot for Stay Cable Maintenance With Security and Rescue Mechanisms	architecture	concept	18.849
(Yu & Chen, 2018)	Failure Mechanisms Cumulative Model for Reliability Evaluation of k-Out-of-n System with Different Types of Failure Mechanisms	anti thesis	anti myopia	19.800
(Sakib & Sun, 2024)	Consolidating Trees of Robotic Plans Generated Using Large Language Models to Improve Reliability	anti thesis	anti myopia	19.799
(Hahnloser et al., 2023)	Feasibility of on-demand robotics in colorectal surgery: first cases	openalex robotics	bulk concept	14.798
(Pool, 2024)	Addressing uncertainty challenges for autonomous driving in real-world environments	deployment	concept	17.754
(Webster et al., 2008)	Mechanics of Precurved-Tube Continuum Robots	openalex robotics	bulk concept	14.700
(Gebhard et al., 2025)	Flow matching for atmospheric retrieval of exoplanets: Where reliability meets adaptive noise levels	architecture	concept	14.699
(Miret & Krishnan, 2024)	Are LLMs Ready for Real-World Materials Discovery?	ebm transformer	concept	17.696
(Acharya & Song, 2026)	A Comprehensive Review of Neuro-symbolic AI for Robustness, Uncertainty Quantification, and Intervenable	architecture	concept	19.690
(Ghadi et al., 2023)	Integration of federated learning with IoT for smart cities applications, challenges, and solutions	ebm	concept	15.689
(Zhang et al., 2023)	Adaptive Safety-Critical Control With Uncertainty Estimation for Human-Robot Collaboration	openalex robotics	bulk concept	15.686
(Bharadwaja et al., 2022)	Physics-Informed Machine Learning and Uncertainty Quantification for Mechanics of Heterogeneous Materials	architecture	concept	15.666
(Fu et al., 2022)	Robust Task Scheduling for Heterogeneous Robot Teams Under Capability Uncertainty	openalex robotics	bulk concept	17.653
(Koizumi et al., 2024)	Performance of uncertainty-based active learning for efficient approximation of black-box functions in materials science	ebm	concept	15.651
(Nardi et al., 2026)	Isolating Error-based and Reward-based Learning Mechanisms in a Real-World Task via Gradual Perturbations in Embodied VR	world models	concept	15.650
(Zhang & Adebayo, 2026)	Open-World Reliability for Vision Foundation Models: A Compliance and Audit Framework for Robustness, Shift Awareness, and Category Discovery	vla	concept	17.650
(Nardi et al., 2025)	Motor learning mechanisms are not modified by feedback manipulations in a real-world task	deployment	concept	16.647
(Wang et al., 2025)	Modeling and Analysis of Pose Accuracy Reliability of Parallel Mechanisms	crossref robotics	bulk concept	14.647
(Betancourt & Muhanna, 2022)	Interval deep learning for computational mechanics problems under input uncertainty	anti thesis	anti myopia	18.640
(Ma et al., 2022)	Sex Robots: Are We Ready for Them? An Exploration of the Psychological Mechanisms Underlying People’s Receptiveness of Sex Robots	openalex robotics	bulk concept	14.618
(Haar & Faisal, 2020)	Brain Activity Reveals Multiple Motor-Learning Mechanisms in a Real-World Task	deployment	concept	16.576

Table 11: Closest local-pool references used to set the novelty boundary. The bright boxed citations jump to bibliography entries. (continued)

Citation	Prior-work threat	Family	Role	Hostile
(Stephens et al., 2024)	Planning under uncertainty for safe robot exploration using Gaussian process prediction	failure	hostile	19.555
(Johnston et al., 2024)	Design Considerations and Robustness to Parameter Uncertainty in Wire-Wrapped Cam Mechanisms	crossref robotics	bulk concept	16.555
(Newbery, 2016)	Missing money and missing markets: Reliability, capacity auctions and interconnectors	ebm	concept	14.551
(Paolino et al., 2013)	Uncertainty in fatigue loading: Consequences on statistical evaluation of reliability in service	anti thesis	anti myopia	19.547
(Yang et al., 2026)	MiraBench: Evaluating Action-Conditioned Reliability in Robotic World Models	wam	concept	17.500
(Lu et al., 2025b)	Toward Real-world BEV Perception: Depth Uncertainty Estimation via Gaussian Splatting	architecture	concept	16.496
(Akhtyamov et al., 2025)	Social robot navigation through constrained optimization: A comprehensive study of uncertainty-based objectives and constraints in the simulated and real world	wam	concept	16.496
(Kim et al., 2019)	Task complexity interacts with state-space uncertainty in the arbitration between model-based and model-free learning	architecture	concept	15.416
(Liao et al., 2026)	Addressing corner cases in autonomous driving: A world model-based approach with mixture of experts and LLMs	architecture	concept	14.405
(Farsi & Zio, 2019)	Industry 4.0: Some Challenges and Opportunities for Reliability Engineering	ebm transformer	concept	15.403
(Rodríguez, 2021)	The unstable queen: Uncertainty, mechanics, and tactile feedback	tactile	concept	15.389
(Minniti et al., 2021)	Adaptive CLF-MPC With Application to Quadrupedal Robots	openalex robotics	bulk concept	16.388
(Ullah et al., 2026)	Applied Mechanics and Materials	openalex robotics	bulk concept	14.350
(Ferraro et al., 2025)	FOCUS: object-centric world models for robotic manipulation	world models	concept	15.343
(Laaouach & Marzouki, 2025)	Physics-Informed Graph Neural Networks (PI-GNN) with Uncertainty Quantification for Optimizing Oil Recovery	architecture	concept	16.343
(Li et al., 2025)	Deep learning predicts real-world electric vehicle direct current charging profiles and durations	deployment	concept	17.343
(Raimondo, 2020)	Getting Practical With Causal Mechanisms: The application of Process-Tracing Under Real-World Evaluation Constraints	architecture	concept	17.340
(Pech et al., 2022)	A hybrid multi-phase field model to describe cohesive failure in orthotropic materials, assessed by modeling failure mechanisms in wood	anti thesis	anti myopia	18.333
(Rahmaty, 2023)	Machine learning with big data to solve real-world problems	deployment	concept	16.317
(Matuszek, 2021)	Grounded Language Learning: Where Robotics and NLP Meet	openalex robotics	bulk concept	14.313
(Abbas, 2026)	Adaptive Governance for a Warming World: Resilience, Fluid Dynamics, and Feedback Mechanisms in Climate Change Mitigation	architecture	concept	14.300
(Ladu et al., 2025)	The Rise of Transformers - Redefining the Landscape of Artificial Intelligence	ebm transformer	concept	14.292

13 DISCUSSION

The result is useful because it clarifies the next research step. Mechanics-gap diagnosis should be coupled to calibrated interventions and external robot validation; otherwise it can improve hidden-mechanism F1 while losing to a robust fallback on the metrics that matter for deployment.

14 REPRODUCIBILITY STATEMENT

All reported rows were generated locally on CPU with deterministic seeds and streaming CSV writes. The source runner is `src/run_experiment.py`; generated evidence is in `results/`; figures are in `figures/`; and this manuscript is generated by `scripts/generate_manuscript.py`. The final PDF is copied only to `Downloads/88.pdf`; no PDF is copied to the visible Desktop.

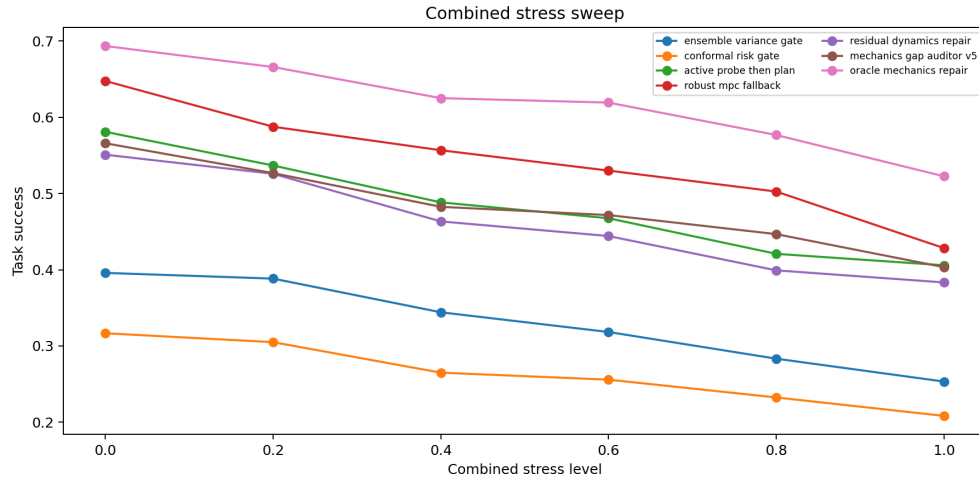


Figure 4: Task success under combined stress.

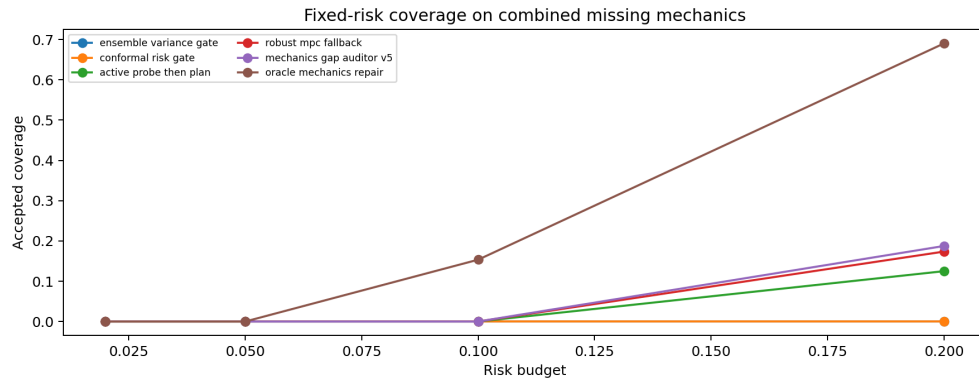


Figure 5: Fixed-risk coverage on combined missing mechanics. Strict coverage collapses.

A APPENDIX: FULL STRESS AND DEPLOYMENT TABLES

The preceding tables are intentionally complete rather than cherry-picked.

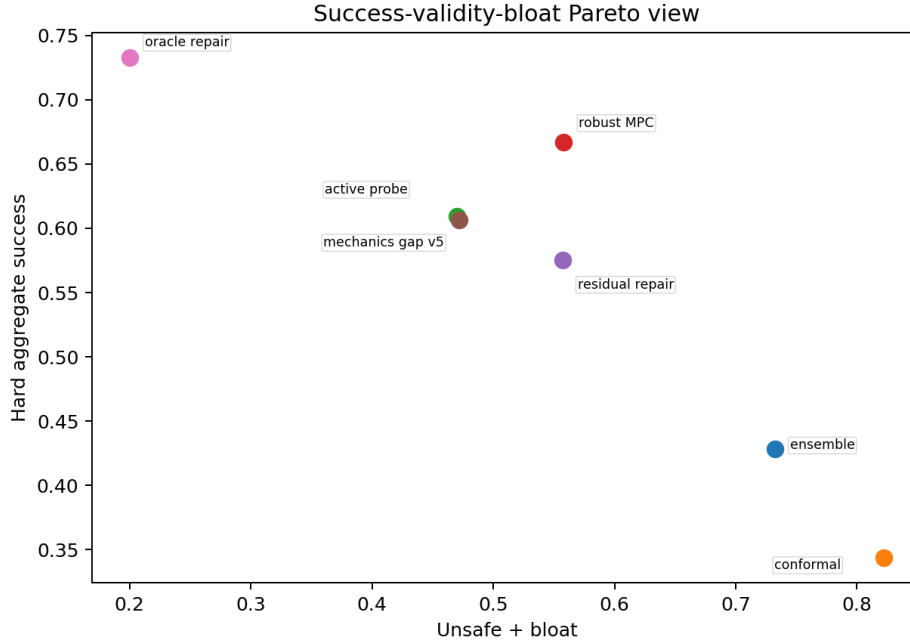


Figure 6: Hard-aggregate Pareto view for success, unsafe action, bloat, and utility.

B APPENDIX: RAW SUMMARY EXTRACT

Table 12: Wrapped extract from results/summary.txt.

Line	Summary extract
1	Paper 88 world_model.uncertainty_bloat v5 expanded audit
2	Terminal recommendation: KILL_ARCHIVE
3	ICLR main ready: no
4	Reason: expanded CPU-only world-model uncertainty-bloat audit adds stronger UQ, conformal, active-probe, robust-MPC, residual-repair, Bayesian-expansion, ablation, stress, and fixed-risk tests, but no real robot or accepted high-f
5	Main rollout rows: 199680
6	Dataset summary rows: 15360
7	Main seed-metric rows: 1040
8	Main metric rows: 1248
9	Main pairwise rows: 672
10	Hard aggregate seed rows: 130
11	Hard aggregate metric rows: 156
12	Hard aggregate pairwise rows: 84
13	Ablation rollout rows: 33600
14	Ablation seed rows: 200
15	Ablation metric rows: 20
16	Stress raw rows: 302400
17	Stress seed rows: 2520
18	Stress metric rows: 252
19	Fixed-risk raw rows: 69120
20	Fixed-risk seed rows: 480
21	Fixed-risk metric rows: 48
22	Fixed-risk pairwise rows: 200
23	Negative cases: 24
25	Frozen hard-aggregate gate:
26	best_success.reference=robust_mpc fallback
27	best_f1.reference=active_probe.then.plan
28	safest.reference=robust_mpc fallback
29	lowest_abstain.reference=robust_mpc fallback
30	lowest_bloat.reference=active_probe.then.plan

Table 12: Wrapped extract from results/summary.txt. (continued)

Line	Summary extract
31	best_utility_reference=robust_mpc_fallback
32	proposal_success=0.60642
33	best_success=0.66701
34	proposal_f1=0.58903
35	best_f1=0.56956
36	proposal_unsafe=0.13768
37	safest_unsafe=0.04861
38	proposal_abstain=0.08003
39	lowest_abstain=0.04792
40	proposal_bloat=0.33433
41	lowest_bloat=0.31471
42	proposal_utility=0.23714
43	best_utility=0.39872
44	paired_success_lower95=-0.07734
45	paired_unsafe_upper95=0.10347
46	main_gate=False
47	mechanism_gate=False
48	stress_gate=True
49	stress_dominated_by=none
50	fixed_risk_gate=False
51	scope_gate=False
52	missing_contact_mode_shift: v5_coverage=0.00000, best_feasible_coverage=0.00000, v5_success=0.00000, best_feasible_success=0.00000, v5_unsafe=0.00000, best_feasible_unsafe=1.00000 — combined_missing_mechanics: v5_coverage=0.00000,
54	Hard aggregate metrics:
55	mean_world_model_mpc task_success=0.51059 ci95=0.01382 f1=0.12245 unsafe=0.19149 false_abstention=0.05747 bloat=0.32166 utility=0.08450
56	ensemble_variance_gate task_success=0.42812 ci95=0.00990 f1=0.40517 unsafe=0.13733 false_abstention=0.24114 bloat=0.59478 utility=-0.14106
57	mc_dropout_uncertainty task_success=0.42899 ci95=0.01726 f1=0.38909 unsafe=0.12344 false_abstention=0.23976 bloat=0.68705 utility=-0.14940
58	conformal_risk_gate task_success=0.34375 ci95=0.00998 f1=0.24982 unsafe=0.05747 false_abstention=0.43906 bloat=0.76475 utility=-0.31438
59	epistemic_ensemble_planner task_success=0.52326 ci95=0.00701 f1=0.44923 unsafe=0.13472 false_abstention=0.13246 bloat=0.51932 utility=0.06403
60	active_probe_then_plan task_success=0.60938 ci95=0.01263 f1=0.56956 unsafe=0.15538 false_abstention=0.04861 bloat=0.31471 utility=0.24197
61	robust_mpc_fallback task_success=0.66701 ci95=0.01149 f1=0.14463 unsafe=0.04861 false_abstention=0.04792 bloat=0.50950 utility=0.39872
62	residual_dynamics_repair task_success=0.57517 ci95=0.01287 f1=0.48802 unsafe=0.15295 false_abstention=0.07309 bloat=0.40408 utility=0.16994
63	bayesian_model_expansion task_success=0.56111 ci95=0.01820 f1=0.54754 unsafe=0.13195 false_abstention=0.11389 bloat=0.41771 utility=0.15082
64	uncertainty_bloat_audit_v4 task_success=0.54080 ci95=0.01434 f1=0.44907 unsafe=0.16701 false_abstention=0.07413 bloat=0.51113 utility=0.08311
65	mechanics_gap_auditor_v5 task_success=0.60642 ci95=0.01538 f1=0.58903 unsafe=0.13768 false_abstention=0.08003 bloat=0.33433 utility=0.23714
66	oracle_mechanics_repair task_success=0.73264 ci95=0.00892 f1=0.72633 unsafe=0.08524 false_abstention=0.04445 bloat=0.11493 utility=0.52000
67	oracle_full_state_policy task_success=0.75851 ci95=0.00925 f1=0.75703 unsafe=0.05799 false_abstention=0.04809 bloat=0.06241 utility=0.59694
69	Key paired hard-aggregate differences:
70	v5_minus_robust_mpc_fallback task_success: mean=-0.06059 ci95=0.01675 lower95=-0.07734 upper95=-0.04384
71	v5_minus_robust_mpc_fallback hidden_mechanism_f1: mean=0.44440 ci95=0.02174 lower95=0.42267 upper95=0.46614
72	v5_minus_robust_mpc_fallback unsafe_action: mean=0.08906 ci95=0.01440 lower95=0.07466 upper95=0.10347
73	v5_minus_robust_mpc_fallback bloat_index: mean=-0.17517 ci95=0.00202 lower95=-0.17719 upper95=-0.17315
74	v5_minus_robust_mpc_fallback robust_utility: mean=-0.16158 ci95=0.03182 lower95=-0.19340 upper95=-0.12976
75	v5_minus_active_probe_then_plan task_success: mean=-0.00295 ci95=0.01759 lower95=-0.02054 upper95=0.01463
76	v5_minus_active_probe_then_plan hidden_mechanism_f1: mean=0.01947 ci95=0.02300 lower95=-0.00353 upper95=0.04247
77	v5_minus_active_probe_then_plan unsafe_action: mean=-0.01771 ci95=0.01567 lower95=-0.03338 upper95=-0.00203
78	v5_minus_active_probe_then_plan bloat_index: mean=0.01962 ci95=0.00246 lower95=0.01716 upper95=0.02209
79	v5_minus_active_probe_then_plan robust_utility: mean=-0.00483 ci95=0.03990 lower95=-0.04473 upper95=0.03507
80	v5_minus_robust_mpc_fallback task_success: mean=-0.06059 ci95=0.01675 lower95=-0.07734 upper95=-0.04384
81	v5_minus_robust_mpc_fallback hidden_mechanism_f1: mean=0.44440 ci95=0.02174 lower95=0.42267 upper95=0.46614
82	v5_minus_robust_mpc_fallback unsafe_action: mean=0.08906 ci95=0.01440 lower95=0.07466 upper95=0.10347
83	v5_minus_robust_mpc_fallback bloat_index: mean=-0.17517 ci95=0.00202 lower95=-0.17719 upper95=-0.17315
84	v5_minus_robust_mpc_fallback robust_utility: mean=-0.16158 ci95=0.03182 lower95=-0.19340 upper95=-0.12976
85	v5_minus_uncertainty_bloat_audit_v4 task_success: mean=0.06562 ci95=0.02270 lower95=0.04292 upper95=0.08833
86	v5_minus_uncertainty_bloat_audit_v4 hidden_mechanism_f1: mean=0.13996 ci95=0.01933 lower95=0.12063 upper95=0.15929
87	v5_minus_uncertainty_bloat_audit_v4 unsafe_action: mean=-0.02934 ci95=0.01703 lower95=-0.04637 upper95=-0.01231
88	v5_minus_uncertainty_bloat_audit_v4 bloat_index: mean=-0.17680 ci95=0.00233 lower95=-0.17913 upper95=-0.17448
89	v5_minus_uncertainty_bloat_audit_v4 robust_utility: mean=0.15403 ci95=0.04678 lower95=0.10725 upper95=0.20081

Table 12: Wrapped extract from results/summary.txt. (continued)

Line	Summary extract
90	v5_minus_robust_mpc_fallback task_success: mean=-0.06059 ci95=0.01675 lower95=-0.07734 upper95=-0.04384
91	v5_minus_robust_mpc_fallback hidden_mechanism.fl: mean=0.44440 ci95=0.02174 lower95=0.42267 upper95=0.46614
92	v5_minus_robust_mpc_fallback unsafe_action: mean=0.08906 ci95=0.01440 lower95=0.07466 upper95=0.10347
93	v5_minus_robust_mpc_fallback bloat_index: mean=-0.17517 ci95=0.00202 lower95=-0.17719 upper95=-0.17315

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